SIADS 696 Report: Predicting Myocardial Infarction

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Introduction

In the United States, someone experiences a myocardial infarction (MI) every 40 seconds, totaling roughly 805,000 events annually. Many high-risk individuals go unidentified until after a cardiac event, and preventive interventions are often applied broadly rather than targeted at those most likely to benefit. Catching these individuals earlier, and understanding which behavioral subgroups they fall into, could cut MI incidence and direct prevention resources where they're actually needed.

Traditional clinical risk assessment relies on lab work, physician visits, and medical records. These methods are costly, intrusive, and inaccessible to large segments of the population. Survey-based approaches are lower-cost, scalable, and non-intrusive, while capturing behavioral and social determinants of health that clinical tools often overlook. That gap is what led us to test whether survey data alone could support population-level cardiovascular screening.

We use the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2024 dataset, which contains self-reported health data from U.S. residents. Our framework combines supervised and unsupervised machine learning: **Logistic Regression**, **Random Forest**, and **XGBoost** for MI risk prediction, and **K-Prototypes clustering** and **Gaussian Mixture Models** (GMM) to identify behavioral risk profiles and estimate intervention benefit. Unlike clinically driven approaches, our pipeline relies entirely on survey data, making it low-cost and broadly deployable.

Random Forest was selected as the final supervised model, achieving an **AUC** of **0.8323** with strong recall, with age, general health, and stroke history emerging as the strongest predictors. **Unsupervised clustering** identified four distinct population subgroups; the highest-risk cluster had a **22.3% MI prevalence**, characterized by severely poor physical health, active smoking history, and multiple chronic conditions, including arthritis and kidney disease. The two components provide a more complete picture of cardiovascular risk, identifying both who is most likely to have a history of MI and what the highest-risk behavioral profiles look like.

Related work

The CMS Million Hearts Model (concluded 2023) used clinical risk scores to identify Medicare beneficiaries for preventive interventions[1]. Our approach instead uses self-reported survey data and connects supervised risk prediction with unsupervised behavioral clustering to support targeted intervention planning.

Rabba et al. (2023) showed that machine learning on CDC survey data can accurately predict heart disease using a reduced question subset[2]. We extend this using BRFSS data and add unsupervised clustering for population-level behavioral profiling beyond prediction alone.
Wu et al. (2025) validated supervised ML models for coronary heart disease prediction using non-invasive BRFSS and NHANES indicators[3], finding LightGBM performed best. Our work

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[1] Centers for Medicare & Medicaid Services. (n.d.). *Million Hearts Cardiovascular Disease Risk Reduction Model*. CMS. https://www.cms.gov/priorities/innovation/innovation-models/million-hearts-cvdrrm
[2] Rababa, S., Yamin, A., Lu, S., & Obaidat, A. (2023). *Predicting Heart Disease and Reducing Survey Time Using Machine Learning Algorithms*. *arXiv*. https://arxiv.org/abs/2306.00023
[3] Wu, B., et al. (2025). Building and validating a machine learning model to predict coronary heart disease risk based on non-invasive indicators. *Computer Methods and Programs in Biomedicine*. https://doi.org/10.1016/j.cmpb.2025.109186

evaluates both supervised and unsupervised methods: Logistic Regression, XGBoost, Random Forest, K-Prototypes, and GMM, targeting both risk prediction and behavioral subgroup discovery.

Data Source(s)

The dataset was sourced from the **CDC Behavioral Risk Factor Surveillance System (BRFSS) 2024**, available for download at https://www.cdc.gov/brfss/annual_data/2024/files/LLCP2024XPT.zip. The data is distributed as a SAS Transport file (.XPT), which was read into Python using the pandas.read_sas() function. The 2024 release reflects survey responses collected during the 2024 calendar year from U.S. adult residents across all 50 states, the District of Columbia, and participating U.S. territories.

The dataset contains the binary target variable of our interest _MICHD, indicating whether a respondent has ever been diagnosed with coronary heart disease or a myocardial infarction- along with variables spanning domains like: demographics, socioeconomic and social determinants, cardiovascular risk factors and comorbidities, mental health and functional limitations, healthcare access and preventive care, adverse childhood experiences, diet, cognitive decline, and survey geography. The base sample consisted of 457,670 records. As a cross-sectional telephone survey, the BRFSS contains no longitudinal time element within a single annual release; all responses reflect a single point-in-time self-report from 2024.

Feature engineering

Initial Filtering

The full BRFSS dataset contains several hundred variables, many of which are redundant recodes, administratively oriented, or too narrowly defined to generalize. We subsetted to 73 variables by removing redundancies and dropping non-actionable features. Records missing the target variable (~5,206) were removed, yielding an analytic sample of 452,464 records.

Survey Code Remapping

BRFSS encodes non-response using sentinel values (e.g., 7, 9, 77, 99) that are numerically meaningful but semantically missing. Retaining them as raw integers would cause models to treat "refused to answer" as a meaningful ordinal value (e.g., a refusal code of 9 interpreted as falling between valid responses 1 and 10). All such values were remapped to NaN or an explicit "Unknown"/"Refused" category depending on variable type. The target variable was remapped from {1, 2} to binary {1, 0}.

Categorical Encoding

Nominal variables were decoded to labeled string categories and cast to `category` dtype to prevent tree-based models from imposing false ordinal relationships and to ensure one-hot encoding maps to meaningful labels. This applies to variables such as race/ethnicity, marital status, insurance type, diabetes status, smoking status, life satisfaction, and loneliness. Binary yes/no variables were mapped to {"Yes", "No", "Unknown"} to preserve the distinction between a "No" response and a missing or refused one, as non-participation patterns may themselves be informative.

Continuous Variable Handling

Sentinel codes in continuous fields (days of poor health, alcohol consumption, smoking exposure) were replaced with NaN and columns retained as float. The value 88 ("none/not applicable") was mapped to 0 where appropriate, as non-consumers represent a true zero; treating them as missing would systematically undercount abstainers and bias estimates.

Derived Features

The sex variable was converted to a binary numeric indicator, as this is more efficient than a two-level categorical for modeling. Sugary drink frequency uses a multi-part format encoding both time unit in the hundreds (1xx) and count in a single integer (x11), making raw codes non-comparable across respondents (e.g., 201 for once per week appears larger than 103 for three times per day). The variable was converted to estimated daily frequency for comparability. Don't know and refused codes were retained as NaN.

Feature Selection

Feature selection was done in three stages: univariate association testing, random forest importance, and inter-feature correlation analysis to catch multicollinearity."

Stage 1: Univariate Association Testing

Features were grouped into categorical (string/object or fewer than 10 unique values) and continuous (numeric with high cardinality), as each type requires different statistical tests. Of the 72 candidate features, 58 were categorical and 14 continuous.

Categorical features were evaluated using chi-square and Cramer's V. Chi-square tests whether a feature's distribution differs across MI/non-MI groups; Cramer's V quantifies effect size on a [0, 1] scale independent of sample size. The latter is critical at ~452k records, where virtually any association reaches statistical significance regardless of practical importance.

Continuous features were evaluated using the Mann-Whitney U test and point-biserial correlation. Mann-Whitney U is non-parametric and assumption-free regarding normality, making it appropriate for the heavily skewed BRFSS variables in this group. Point-biserial correlation serves as the corresponding effect size measure.

Table 1. Feature Selection Classification Criteria

Category	Criteria
both	$p \leq 0.05$ and effect size ≥ threshold (retained)
p_only	$p \leq 0.05$ but weak effect (excluded)
cv_only	Effect size above threshold but unreliable p (excluded)
neither	Neither criterion met (excluded)

Effect size thresholds were set at Cramer's V >= 0.09 for categorical features and $|r| > 0.10$ for continuous features. Features classified as both were retained, yielding 27 features (21 categorical, 6 continuous) ranked by association strength.

Stage 2: Random Forest Interaction-Aware Cross-Check

Univariate tests evaluate features in isolation and can miss interaction effects. For example, BMI alone may be weakly correlated with MI, but combined with age and diabetes it may compound risk. To capture these effects, a Random Forest (500 trees, min_samples_leaf = 20) was trained on all 72 features using random undersampling to address class imbalance.

Features with importance above the baseline threshold of 1/72 (~0.014) were compared against the univariate selection. Of 17 features ranked above baseline by the RF, 16 overlapped with the univariate set. Eleven univariate-selected features fell below the RF baseline, suggesting strong individual signal but limited interaction value. One feature, sex, was identified by the RF (importance = 0.032) but excluded by univariate testing (Cramer's V = 0.067, below the 0.09 threshold), and was added to the final set, bringing the combined selection to 28 features.

Stage 3: Inter-Feature Correlation

To detect multicollinearity among the 28 selected features, pairwise correlation was computed using the metric appropriate to each pair type: Cramer's V for categorical-categorical pairs, Pearson r for continuous-continuous pairs, and Cramer's V (after binning the continuous variable into five ordinal categories) for categorical-continuous pairs. Pairs exceeding a correlation threshold of 0.7 were flagged as highly collinear. No pairs exceeded this threshold, confirming that the final feature set does not contain redundant features that would destabilize model coefficients or distort feature importance scores.

Part A. Supervised Learning

Methods Description

Three supervised learning models were evaluated: Logistic Regression, Random Forest, and XGBoost. All three were trained on the same preprocessing pipeline and evaluated with five-fold cross-validation, Logistic Regression as an interpretable baseline, Random Forest and XGBoost as tree-based ensembles capable of capturing nonlinear interactions. The best-performing model was subsequently analyzed using SHAP, feature ablation, sensitivity analysis, and failure analysis to understand its predictive behavior and limitations.

Logistic Regression served as the baseline model because it provides an interpretable probabilistic framework for binary classification and estimates the probability of myocardial infarction (MI) for each respondent. Random Forest was selected as a tree-based bagging ensemble method that combines predictions from multiple trees trained on bootstrap samples and random subsets of features. By aggregating predictions across many independent trees, Random Forest reduces variance, captures nonlinear relationships and feature interactions, and performs well on heterogeneous tabular datasets such as BRFSS[4]. XGBoost was included as it is a gradient boosting ensemble that builds decision trees sequentially, with each tree learning to correct the errors of the previous ensemble. By optimizing a regularized objective function and efficiently handling large-scale datasets, XGBoost produces accurate and computationally efficient predictive models[5].

After comparing the three candidate models, Random Forest was selected for further optimization because it achieved the strongest overall performance while maximizing recall for the myocardial infarction (MI) class. Hyperparameter tuning focused on the min_samples_leaf parameter, which controls tree complexity. Candidate values of 1, 5, 10, 20, 50, and 100 were evaluated using five-fold cross-validation, with a value of 20 selected because it achieved performance comparable to the best-performing configurations while producing a simpler, more robust model. The effect of this parameter is examined further in the Sensitivity Analysis section.

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[4] Breiman, L. (2001). *Random Forests*. *Machine Learning, 45*(1), 5-32. https://doi.org/10.1023/A:1010933404324
[5] Chen, T., & Guestrin, C. (2016). *XGBoost: A scalable tree boosting system*. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785-794). https://doi.org/10.1145/2939672.2939785

Feature Representation

As described in the Feature Engineering section, the supervised learning dataset consisted of 27 selected predictor variables representing demographic, behavioral, socioeconomic, and health-related characteristics. These variables included both categorical and continuous features and were processed using a common preprocessing pipeline built with a ColumnTransformer:

- **Categorical features** were imputed using the most frequent category and one-hot encoded to avoid introducing an artificial ordinal relationship between categories.

- **Continuous features** were imputed using the median and standardized to place variables on a common scale. Standardization was particularly important for Logistic Regression, whose coefficients are sensitive to differences in feature scale.

Supervised Model Evaluation

Because only approximately 9% of respondents reported a history of myocardial infarction (MI), the dataset was highly imbalanced. Under these conditions, overall accuracy can be misleading because a model may achieve high accuracy by simply predicting the majority class while failing to identify respondents with MI.

The primary objective of this study was to identify as many respondents with a history of MI as possible. Consequently, **recall for the MI class** was the primary evaluation metric, as it measures the proportion of true MI cases correctly identified by the model. Emphasizing recall reduces false negatives, which is particularly important in a screening setting where failing to identify an individual with a history of MI is generally more consequential than generating additional false positives.

Recall was complemented by ROC-AUC and Average Precision (AP). **ROC-AUC** measures each model's ability to discriminate between respondents with and without MI across all classification thresholds, while **AP** summarizes performance on the precision-recall curve and is particularly informative for imbalanced datasets. These metrics provide a comprehensive assessment of each model's ability to identify MI cases while maintaining strong overall discrimination.

The three candidate models were evaluated using five-fold cross-validation and an independent test set. Their performance is summarized in **Table 2**.

Table 2. Supervised Learning Models Performance Comparison

Model	CV AUC (± SD)	Test AUC	Average Precision	MI Recall
Random Forest	0.8313 ± 0.0040	0.8323	0.3481	0.80
Logistic Regression	0.8299 ± 0.0039	0.8306	0.3512	0.77
XGBoost	0.8264 ± 0.0035	0.8277	0.3294	0.78

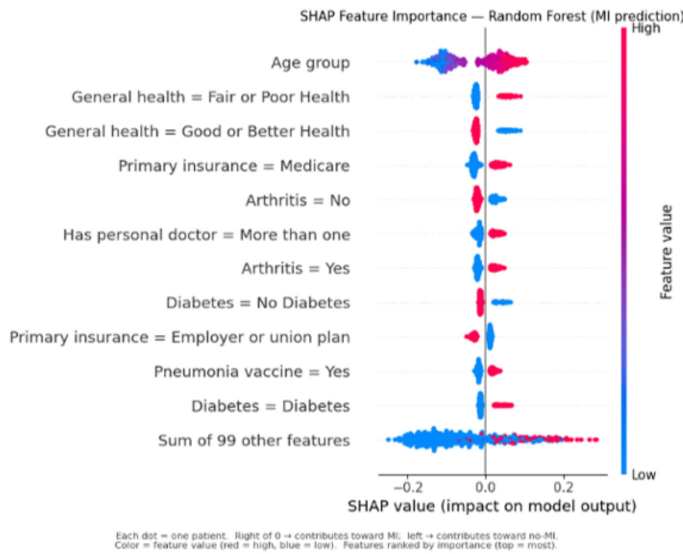
Random Forest achieved the highest recall (0.80) and the highest cross-validated and test ROC-AUC among the three models, demonstrating the strongest overall performance. Although Logistic Regression produced a slightly higher Average Precision, the improvement was marginal and came at the expense of lower recall. Because minimizing missed MI cases was the primary objective, Random Forest was selected for all subsequent analyses.

In-Depth Evaluation of the Random Forest Classifier

Feature and Ablation Analysis

In order to understand how the Random Forest made its predictions, we examined feature importance using two complementary approaches. **SHAP** (SHapley Additive exPlanations) (**Figure 1**) was used to quantify how individual features influenced predictions, while **single-feature ablation** evaluated how much overall model performance changed when each feature was removed (Table 3). These analyses distinguish features that strongly influence individual predictions from those that contribute unique predictive information to the model.

Figure 1: SHAP Feature Importance



The SHAP analysis identified **age group** (_AGEG5YR) as the strongest predictor of myocardial infarction (MI), with older respondents shifting predictions toward the MI class and younger respondents toward the no-MI class. General health was the second most influential predictor, with fair or poor health increasing predicted risk and good or excellent health reducing it.

Beyond these dominant predictors, the model relied on a combination of demographic, clinical, and healthcare utilization variables. Diabetes, arthritis, Medicare coverage, pneumonia vaccination, and having a personal doctor all contributed to higher predicted risk. The broad distribution of SHAP values indicates that many features influenced predictions differently across respondents' overall clinical profiles.

In addition to the SHAP analysis, we performed single-feature ablation by removing one feature at a time, retraining the model, and measuring the change in test AUC. Features that caused larger drops in performance contributed more to the model's predictions.

Table 3. Single-feature Ablation Results

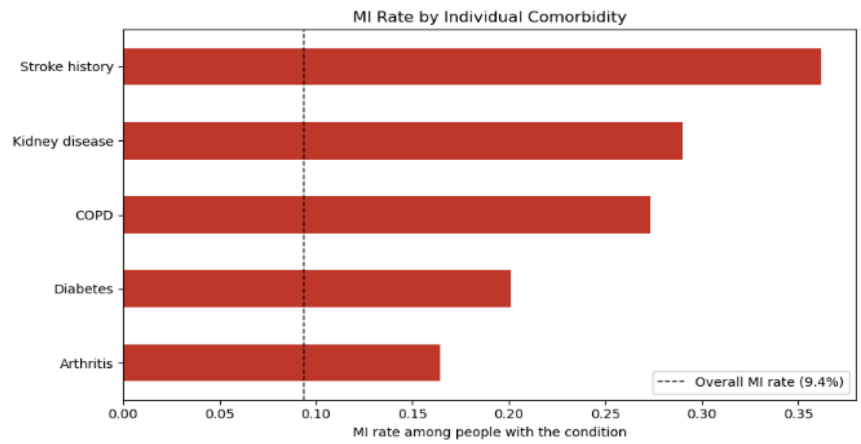
Removed Feature	CV AUC	Test AUC	Δ Test AUC
Baseline (All Features)	0.8313	0.8323	-
CVDSTRK3 (Stroke History)	0.8248	0.8255	−0.0068
_AGEG5YR (Age Group)	0.8255	0.8268	−0.0055
_RFHLTH (General Health)	0.8267	0.8272	−0.0051
PERSDOC3 (Personal Doctor)	0.8277	0.8286	−0.0037
DIABETE4 (Diabetes)	0.8287	0.8298	−0.0025

Although SHAP identified age as the most influential predictor, ablation analysis showed that removing stroke history produced the largest drop in model performance. This difference reflects the distinct purposes of the two methods: SHAP explains individual predictions, whereas ablation measures a feature's overall contribution to model performance.

Observed MI Prevalence Across Major Comorbidities

To further investigate the supervised learning results, we examined the observed prevalence of myocardial infarction among respondents with major comorbidities. Stroke history showed the highest prevalence (~36%), followed by kidney disease (~29%), COPD (~27%), diabetes (~20%), and arthritis (~16%) This lines up with what SHAP and ablation already showed.

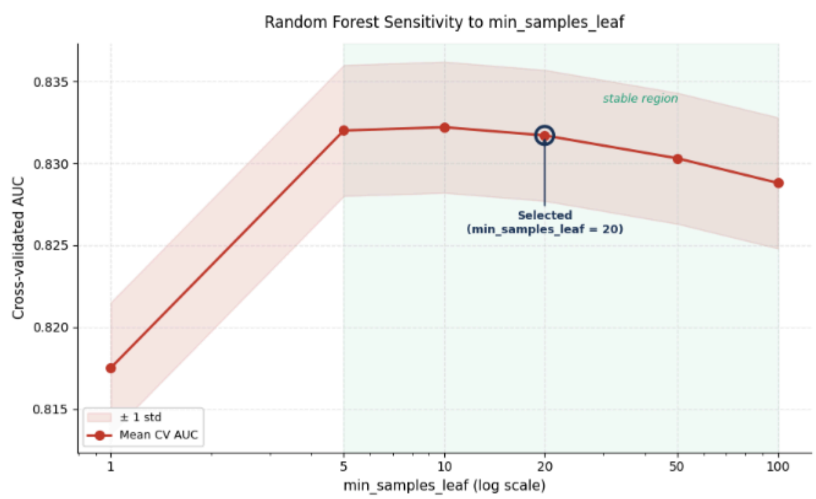
Figure 2: MI Rate by Condition



Sensitivity Analysis

Sensitivity analysis was performed to evaluate how changes in the Random Forest hyperparameter min_samples_leaf affected model performance. This parameter specifies the minimum number of training observations required in each terminal leaf and controls the complexity of the decision trees. Larger values produce simpler trees with lower variance, whereas smaller values allow more complex trees that may overfit the training data.

Figure 3. Sensitivity to *min_samples_leaf*



Cross-validated ROC-AUC remained remarkably stable across a broad range of values, varying only slightly between 0.8302 and 0.8316 for min_samples_leaf values of 5 to 50. Performance declined substantially only when min_samples_leaf = 1, where the mean cross-validated AUC decreased to 0.8137, consistent with the model overfitting by creating overly specific decision trees. At the opposite extreme (100), performance decreased only modestly, suggesting a slight loss of predictive power from excessive smoothing.

The selected value of 20 lies within this stable performance region, indicating that the model is not highly sensitive to moderate changes in this hyperparameter. This robustness increases confidence that the observed performance reflects genuine learning from the data rather than reliance on a narrowly tuned parameter setting.

Tradeoffs

Two tradeoffs shaped the final Random Forest model, both stemming from the goal of catching as many MI cases as possible.

The first tradeoff was **recall versus precision**. Training on an undersampled dataset increased the model's ability to identify true MI cases **(recall = 0.80)** but also increased the number of false positives, resulting in lower precision. In a screening context, this tradeoff is appropriate because failing to identify an individual with a history of MI is generally considered more consequential than recommending additional follow-up for someone who ultimately does not have the condition.

The second tradeoff was predictive performance versus interpretability. Although Logistic Regression achieved similar overall discrimination, Random Forest provided slightly higher recall while capturing nonlinear relationships and feature interactions that a linear model cannot represent. This increased complexity comes at the cost of reduced interpretability; however, SHAP analysis helped address this limitation by explaining how individual features contributed to the model's predictions.

Failure analysis

To review the limitations of the Random Forest classifier, we examined three representative misclassified test records, each illustrating a different failure mechanism. The error-rate-by-age analysis (Figure 3) shows a clear asymmetry in model performance: false negatives remained consistently low (approximately 2%) across all age groups, whereas false positives increased sharply with age, rising from less than 1% among respondents aged 18-24 to more than 60% among those aged 80 years and older.

Figure 4. Error Rates by Age

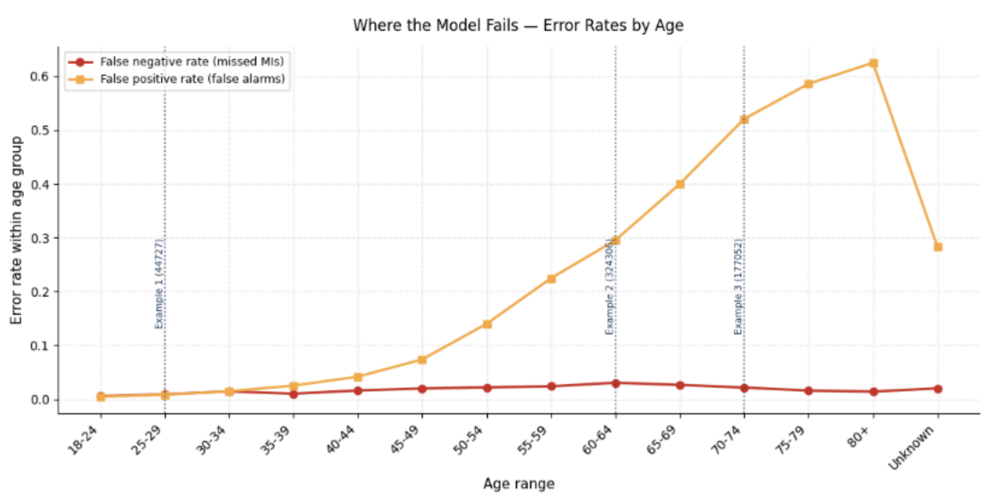


Table 4. Examples of Supervised Model Limitations

Example	Prediction	Prob. MI	Failure Type	Category	Mechanism
1 (44727)	MI=1 Pred=0	0.028	False Negative	Edge case	Feature coverage gap
2 (324306)	MI=0 Pred=1	0.956	False Positive	Systematic	Base-rate inflation
3 (177052)	MI=0 Pred=1	0.500	False Positive	Boundary	Conflicting signals/uncertainty

The three representative cases illustrate different model limitations. Example 1 shows an edge case where limited risk information led to a false negative. Example 2 is a systematic false positive caused by undersampling, while Example 3 is a boundary case in which competing risk factors resulted in an uncertain prediction near the decision threshold.

These failure modes suggest several opportunities for improvement. Richer clinical information, such as family history, laboratory test results, and genetic risk factors, could reduce false negatives. Class weighting, probability calibration, or a more conservative decision threshold could reduce false positives, while borderline cases could be flagged for additional clinical review instead of receiving a binary prediction.These improvements address limitations in the data, model training, and prediction strategy.

Part B. Unsupervised Learning

The goal of this project's unsupervised learning component was to identify distinct population subgroups within the BRFSS dataset with differing susceptibility to myocardial infarction (_MICHD). Two distinct clustering methods were applied: K-Prototypes[6], a non-probabilistic hard-assignment method, and Gaussian Mixture Models (GMM)[7], a probabilistic soft-assignment method.

Missing Value Treatment and Preprocessing

Missing values were addressed before clustering. Continuous variables were imputed using the median, which is robust to outliers and skewed distributions commonly found in health survey data. Categorical variables were imputed using the mode, preserving the most frequently observed category within each feature. All categorical columns were cast to string type for compatibility with Factor Analysis of MIxed Data (FAMD). A stratified sample of 50,000 records was drawn to reduce computational cost for model fitting, preserving the original MICHD class ratio of 9.4%.

Of the 27 selected features, 21 were identified as categorical and 6 as continuous. Classification was determined by dtype and cardinality. Any feature with an object or category dtype, or with 10 or fewer unique values, was treated as categorical. The remaining numeric features were treated as continuous.

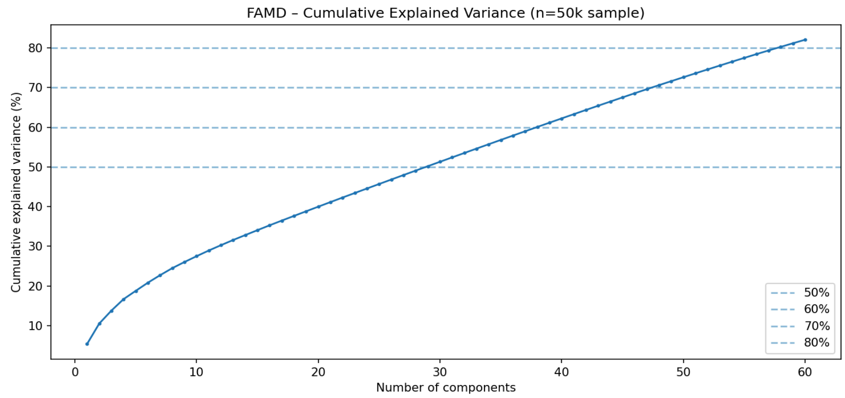
[6] Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery, 2*(3), 283-304. https://doi.org/10.1023/A:1009769707641
[7] Reynolds, D. A. (2009). Gaussian mixture models. In S. Z. Li & A. K. Jain (Eds.), *Encyclopedia of biometrics* (pp. 659-663). Springer. https://doi.org/10.1007/97The remainingMixed8-0-387-73003-5_196

Dimensionality Reduction

FAMD was chosen as the dimensionality reduction method because the BRFSS dataset contains both categorical and continuous features, making standard PCA inadequate. FAMD generalizes PCA by applying PCA to continuous variables and MCA (Multiple Correspondence Analysis) to categorical variables simultaneously, producing a low-dimensional embedding suitable for downstream clustering and visualization[8].

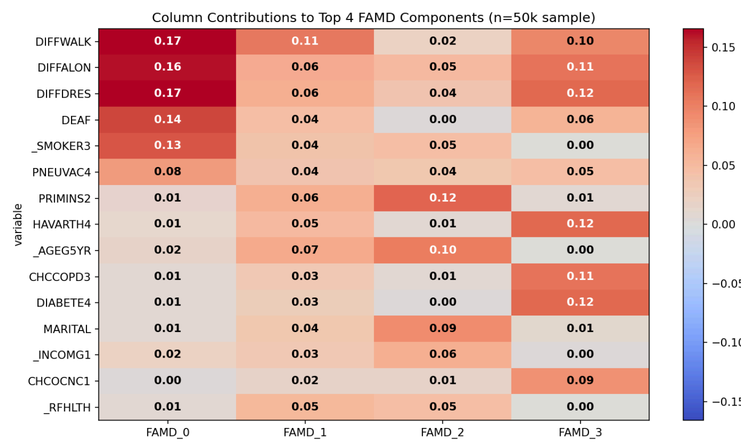
To determine the optimal number of components, FAMD was first fit on the 50k stratified sample with 60 components as an upper bound (Figure 5). Cumulative explained variance was evaluated at four thresholds: 50%, 60%, 70%, and 80%, corresponding to 30, 39, 48, and 58 components. An 80% threshold was selected, retaining 58 components as the final embedding.

Figure 5: FAMD Cumulative Explained Variance



The first four FAMD components captured distinct health and demographic dimensions (**Figure 6**). Component 1 was dominated by functional disability indicators (DIFFWALK, DIFFDRES, DIFFALON) alongside smoking status. Component 2 added age, general health, physical health days, and insurance status. Component 3 captured socioeconomic and demographic factors, including income, marital status, and insurance type. Component 4 loaded heavily on chronic disease comorbidities: diabetes, arthritis, COPD, and kidney disease, which are all clinically meaningful risk factors for MI.

Figure 6: FAMD Component Heatmap



Method 1: K-Prototypes

K-Prototypes was chosen because it handles mixed data types directly, combining Euclidean distance for continuous features with Hamming distance for categorical ones. Standard K-Means is unsuitable for this dataset as Euclidean distance is not meaningful for nominal variables. The algorithm was applied directly to the original feature space, with continuous features standardized using StandardScaler. Due to the computational cost of K-Prototypes on large datasets, the model was fit on the 50k stratified sample, and cluster assignments were then predicted for all 452,464 records.

[8] Pagès, J. (2004). Multiple Factor Analysis: Main Features and Application to Sensory Data.

To determine the optimal number of clusters, K-Prototypes was fit across k=2 to k=8 using the Huang initialization with 3 random restarts per k (**Figure 7**).

Three complementary metrics were evaluated: cost/inertia with marginal gains to identify the elbow, silhouette score, and Davies-Bouldin score.

Figure 7: K-Prototypes Elbow Curve

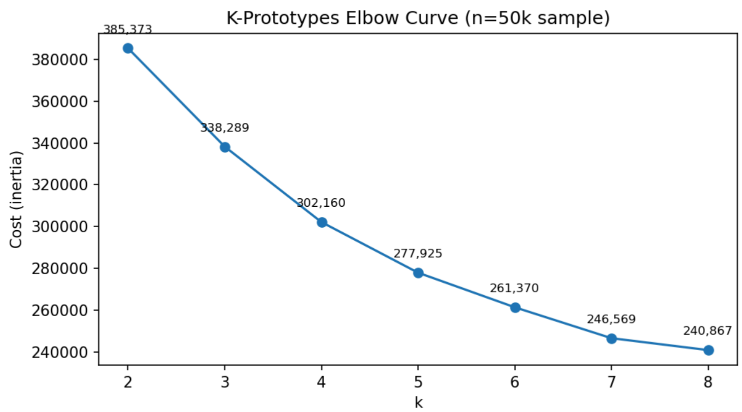
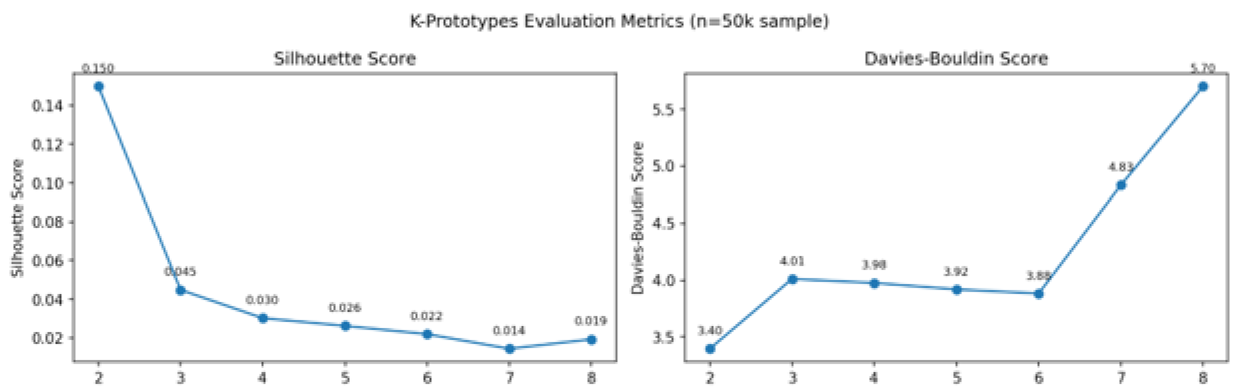


Figure 8: K-Prototypes Metrics



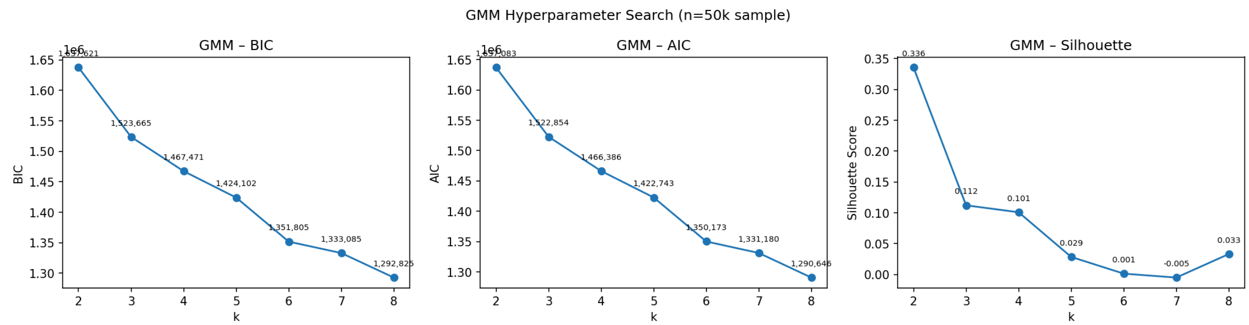
The elbow curve showed the sharpest marginal gain drop between k=4 and k=5, where improvement decreased by approximately 33%. Silhouette scores declined steadily from 0.150 at k=2 to 0.019 at k=8, while Davies-Bouldin scores spiked at k=7 and k=8, indicating instability. Although silhouette peaked at k=2, a two-cluster solution lacks the granularity needed for meaningful health risk profiling. K=4 was selected based on the convergence of marginal gains, stable Davies-Bouldin scores, and clinical interpretability.

Method 2: Gaussian Mixture Models (GMM)

GMM was chosen as a contrast to K-Prototypes since it models data as a mixture of Gaussian distributions rather than forcing hard cluster boundaries, which is closer to how health risk actually behaves, on a continuum rather than in discrete buckets. GMM was applied to the first 15 standardized FAMD components rather than the full 58-dimensional embedding, as high-dimensional inputs cause GMM's covariance estimates to degenerate. Diagonal covariance was used for stability.

BIC (Bayesian Information Criterion), AIC (Akaike Information Criterion), and silhouette scores were evaluated across k=2 to k=8 (**Figure 9**). BIC and AIC both decreased steadily, with the largest drop between k=2 and k=3. Silhouette peaked at k=2 (0.336) then dropped sharply, turning negative at k=4, indicating cluster overlap at higher k. Based on the BIC/AIC elbow and silhouette stability, k=3 was selected.

Figure 9: GMM Evaluation

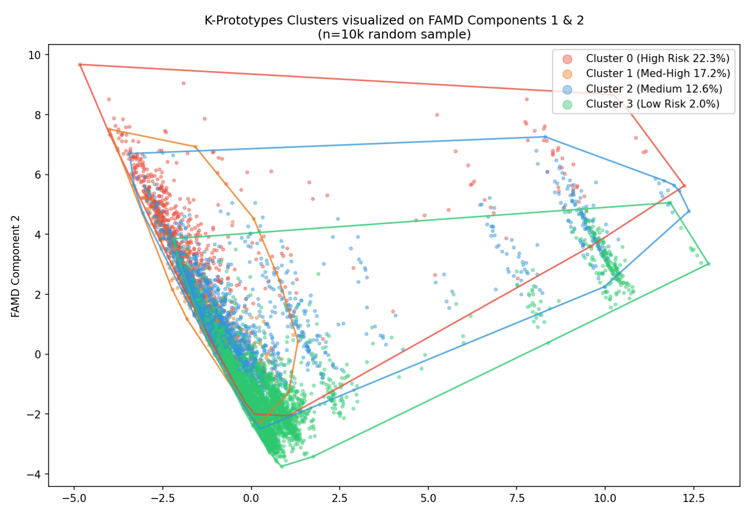


Unsupervised Evaluation

Three metrics were used to evaluate and compare both methods. Silhouette score measures how similar points are to their own cluster relative to other clusters (higher is better). Davies-Bouldin score measures the ratio of within-cluster spread to between-cluster separation (lower is better). Both metrics were computed on the FAMD embedding for comparability. BIC and AIC, which measure statistical fit penalized by model complexity, were used exclusively for GMM hyperparameter selection. Most importantly, MI rate spread across clusters was used as the primary validation metric, as it directly measures whether the clustering solution identifies clinically meaningful risk stratification.

The final K-Prototypes model assigned all 452,464 records across four clusters with substantially different MI rates: Cluster 0 (48,089 members, 22.3% MI), Cluster 1 (36,122 members, 17.1% MI), Cluster 2 (172,795 members, 12.6% MI), and Cluster 3 (195,458 members, 1.8% MI). The 20.5 percentage point spread confirms strong clinical risk stratification.

Figure 10: K-Prototype Clusters

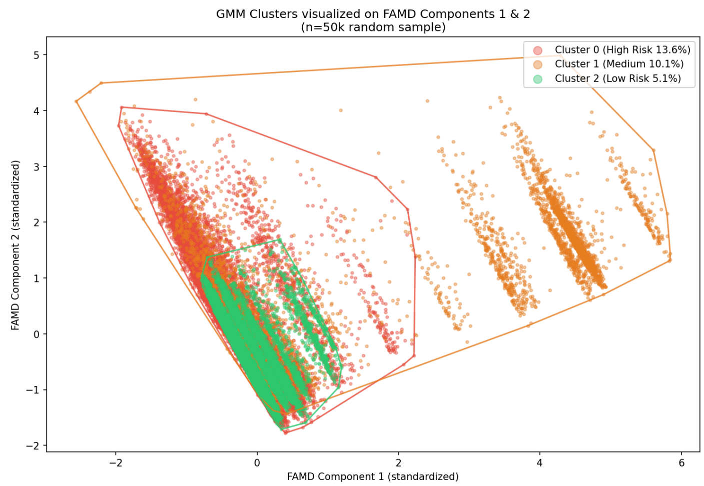


Cluster 0 (highest risk, 22.3%) is defined by severely elevated poor health days (PHYSHLTH +2.40 SD, POORHLTH +2.00 SD), active smoking, arthritis, and multiple doctors. Cluster 1 (17.1%) represents elderly long-term ex-smokers, with the longest time since quitting (_LCSYQTS +2.61 SD), oldest age, and Medicare insurance. Cluster 2 (12.6%) is defined primarily by older age with Medicare coverage. Cluster 3 (1.8%) is the youngest and healthiest group with no distinguishing categorical features, confirming it as the baseline low-risk population.

The final GMM model assigned all 452,464 records across three clusters: Cluster 0 (47,134 members, 11.8% MI), Cluster 1 (260,139 members, 12.3% MI), and Cluster 2 (145,191 members, 3.3% MI). Mean assignment probabilities ranged from 0.959 to 0.981, confirming soft probabilistic assignments. The MIrate spread of 9.0 percentage points (3.3% to 12.3%) confirms meaningful but narrower risk stratification compared to K-Prototypes.

Cluster 2 (3.3% MI) is the most clearly defined: youngest, healthiest, fewest poor health days, and highest income. Clusters 0 and 1 carry similar MI rates (~12%), with Cluster 1 being slightly older with Medicare insurance and Cluster 0 showing marginally worse current health and more missing income data.

Figure 11: GMM Clusters



Method Comparison

K-Prototypes outperformed GMM on every metric relevant to this task (**Table 5**). GMM's negative silhouette score and higher Davies-Bouldin score indicate weaker geometric cluster cohesion, and its MI spread of 9.0 percentage points is less than half of K-Prototypes' 20.5 points. K-Prototypes' ability to operate directly on the original mixed-type feature space, rather than a compressed FAMD embedding, allowed it to capture more clinically meaningful population subgroups. Therefore, K-Prototypes is the recommended cluster assignment model.

Table 5: Unsupervised Model Comparison

Metric	K-Prototypes	GMM
Silhouette	0.029	-0.013
Davies-Bouldin	3.944	4.390
BIC	N/A	1,389,809
AIC	N/A	1,388,998
MICHD Rate Spread	0.205	0.090

Sensitivity Analysis

Sensitivity analysis was performed on K-Prototypes to evaluate robustness to hyperparameter choices and feature representation. Testing across three random seeds revealed minimal sensitivity, with two of three seeds producing identical MICHD spreads of 0.205 and seed 0 showing only a minor deviation of 0.194 attributable to an occasional suboptimal local minimum. The number of initializations showed no sensitivity at all, with n_init values of 1, 3, and 5 all producing identical results, confirming that the solution converges reliably to the same global minimum. The most consequential varying element was feature representation. When GMM was applied to a compressed 15-dimensional FAMD embedding rather than the full original mixed-type feature space used by K-Prototypes, the MICHD spread dropped substantially from 20.5% to 9.0%, demonstrating that input feature representation has a far greater impact on clinical risk stratification than initialization hyperparameters alone

Discussion

Part A- Supervised Learning

Model choice mattered less than expected. All three models produced similar AUC scores, which points to feature availability as the main constraint rather than model complexity. Once Random Forest was selected, the more interesting work was understanding how it made decisions. SHAP and ablation both flagged age, general health, and stroke history as dominant signals, which tracks with established clinical knowledge.

The failure analysis was telling. False negatives stayed low across age groups, but false positives spiked among older respondents, likely because high comorbidity burden in that group gets conflated with MI history. That's a data limitation more than a model one. Family history or lab values would probably help here.

Part B- Unsupervised Learning

The gap between K-Prototypes and GMM was bigger than anticipated. GMM seemed reasonable going in. Health risk is continuous, not discrete, but the MI spread dropped from 20.5 to 9.0 percentage points and the clusters were harder to interpret. Feature representation was the culprit; GMM needed a compressed FAMD embedding while K-Prototypes worked directly on the mixed-type feature space. Algorithm choice matters less than how the data goes in.

The cluster profiles also challenged a simple age-equals-risk assumption. The oldest cluster carried lower MI risk than a younger one with worse current health and active smoking. Cessation history appears more protective than age alone would suggest.

FAMD's scree plot had no clear elbow, which made component selection feel like a judgment call. That was the most uncomfortable part of the pipeline.

Ethical Considerations

Part A- Supervised Learning

BRFSS is self-reported, so prediction quality depends on who responds and how accurately. Non-response tends to be higher in lower-income and uninsured populations, the same groups at elevated cardiovascular risk. The model may perform worst exactly where accurate screening matters most.

The false positive rate climbing with age is a real deployment concern. Older adults without MI history would be disproportionately flagged, which means unnecessary follow-up costs and patient anxiety. This is a population-level screening tool, not a diagnostic one, and should be treated accordingly.

Part B- Unsupervised Learning

Cluster labels shouldn’t be considered a ground truth- they’re our interpretations. Calling a cluster high-risk based on smoking and poor health is reasonable, but those features also correlate with socioeconomic disadvantage. Interventions built on these clusters could end up targeting poverty more than modifiable behavior.

Smaller demographic subgroups may also have been absorbed into larger clusters during fitting, making their distinct profiles invisible. Before using these clusters to allocate resources, MI rates broken down by race, income, and geography should be checked to confirm the segmentation actually holds.

Statement of Work

Alex Lee: Led data import, preprocessing, feature engineering, univariate & multivariate feature selection, code refactoring & cleanup, git repository management, and writing report sections.
Mubashar Khan: Led unsupervised learning by implementing FAMD, K-Prototypes, and GMM clustering, performing tuning and cluster analysis, creating visualizations, and writing report sections.
Maria Febus: Led supervised learning using Logistic Regression, Random Forest, and XGBoost; conducted model evaluation and interpretation, developed visualizations, modularized code, and wrote report sections.

References

1. Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5-32. https://doi.org/10.1023/A:1010933404324
2. Centers for Disease Control and Prevention. (2024, October 24). *Heart disease facts*. https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html
3. Centers for Medicare & Medicaid Services. (n.d.). *Million Hearts Cardiovascular Disease Risk Reduction Model*. https://www.cms.gov/priorities/innovation/innovation-models/million-hearts-cvdrrm
4. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785-794. https://doi.org/10.1145/2939672.2939785
5. Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery, 2*(3), 283-304. https://doi.org/10.1023/A:1009769707641
6. Pagès, J. (2004). Multiple factor analysis: Main features and application to sensory data. *Revue de Statistique Appliquee, 52*(2), 93-111.
7. Rababa, S., Yamin, A., Lu, S., & Obaidat, A. (2023). *Predicting heart disease and reducing survey time using machine learning algorithms*. arXiv. https://arxiv.org/abs/2306.00023
8. Reynolds, D. A. (2009). Gaussian mixture models. In S. Z. Li & A. K. Jain (Eds.), *Encyclopedia of biometrics* (pp. 659-663). Springer. https://doi.org/10.1007/978-0-387-73003-5_196
9. Wu, B., et al. (2025). Building and validating a machine learning model to predict coronary heart disease risk based on non-invasive indicators. *Computer Methods and Programs in Biomedicine*. https://doi.org/10.1016/j.cmpb.2025.109186

Acknowledgement

The authors utilized generative AI tools to assist in prototyping code. To maintain the integrity of this study, all AI-generated content underwent comprehensive human evaluation and verification.

Appendix A: Initial Column List (73 Total)

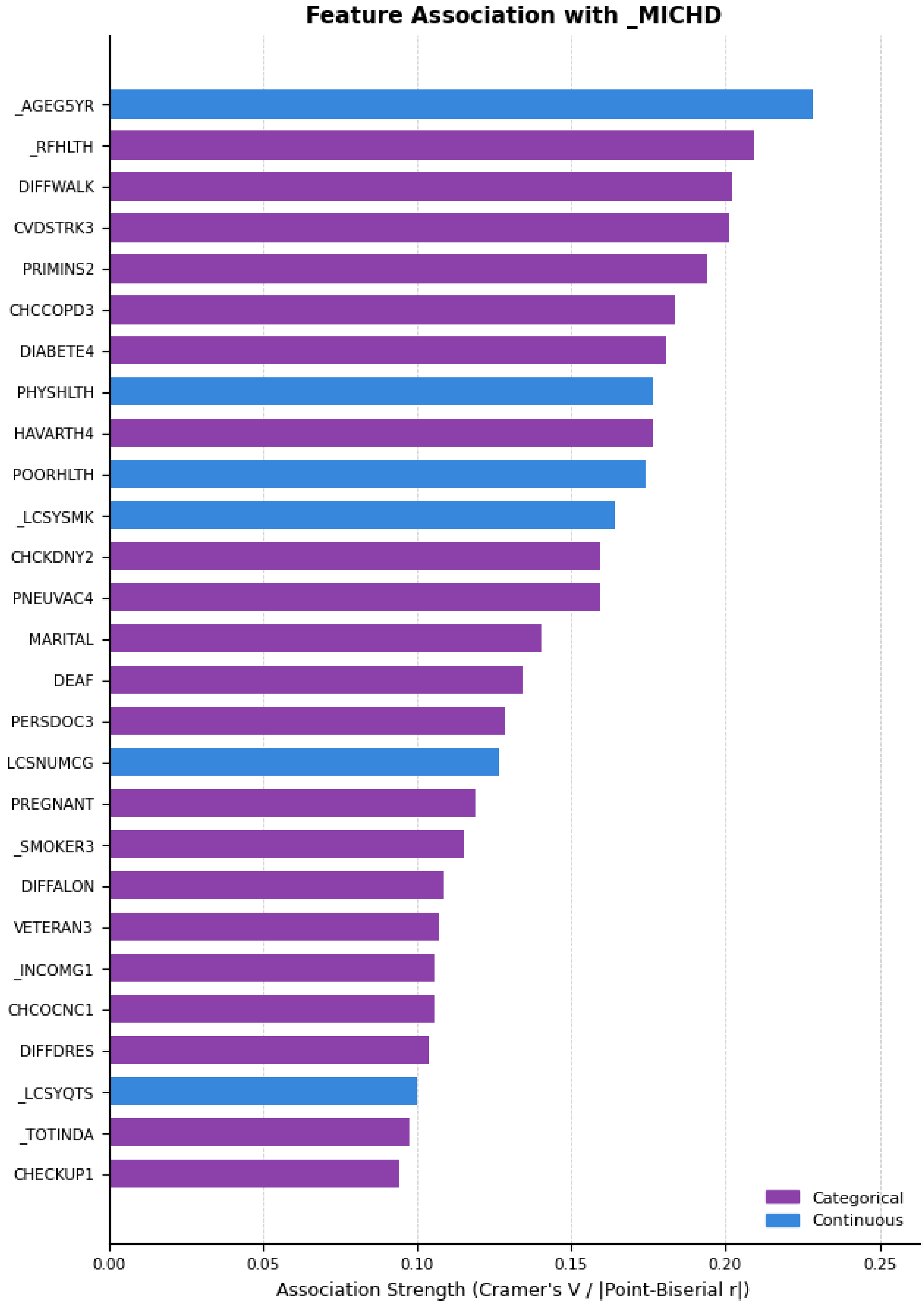
Feature	Description	Type
_MICHD	Target: CHD or MI diagnosis	Binary
_TOTINDA	Treatment: leisure-time physical activity	Binary
_ismale	Sex (male = 1)	Binary
_AGEG5YR	Age in 5-year bands	Ordinal
_RACE	Race/ethnicity	Nominal
MARITAL	Marital status	Nominal
VETERAN3	Veteran status	Nominal
PREGNANT	Pregnancy status	Nominal
_EDUCAG	Education level	Ordinal
_INCOMG1	Income level	Ordinal
SDHBILLS	Unable to pay bills	Nominal
SDHTRNSP	Transportation barriers	Nominal
HOWSAFE1	Perceived neighborhood safety	Ordinal
PRIMINS2	Primary insurance type	Nominal
_RFHLTH	General health status	Nominal
PHYSHLTH	Days physical health not good	Continuous
MENTHLTH	Days mental health not good	Continuous
POORHLTH	Days poor health limited activity	Continuous
CVDSTRK3	History of stroke	Nominal
_RFBMI5	Overweight/obese flag	Nominal
DIABETE4	Diabetes diagnosis	Nominal
DIABTYPE	Diabetes type	Nominal

Feature	Description	Type
INSULIN1	Currently on insulin	Nominal
FEETSORE	Foot sores (diabetes complication)	Nominal
_SMOKER3	Smoking status (4-level)	Nominal
USENOW3	Smokeless tobacco use	Nominal
_CURECI3	Current e-cigarette user	Nominal
LCSFIRST	Age started smoking	Continuous
LCSNUMCG	Avg cigarettes per day	Continuous
_LCSYSMK	Years smoked	Continuous
_LCSYQTS	Years since quitting	Continuous
STOPSMK2	Quit attempt in past 12 months	Nominal
DRNKANY6	Any drinking past 30 days	Nominal
AVEDRNK4	Avg drinks per drinking day	Continuous
_RFBING6	Binge drinker flag	Nominal
_RFDRHV9	Heavy drinker flag	Nominal
_DRNKWK3	Total drinks per week	Continuous
MAXDRNKS	Max drinks on single occasion	Continuous
CHCCOPD3	COPD/emphysema	Nominal
CHCKDNY2	Kidney disease	Nominal
HAVARTH4	Arthritis	Nominal
_LTASTH1	Ever had asthma	Nominal
ADDEPEV3	Depressive disorder	Nominal
CHCSCNC1	Non-melanoma skin cancer	Nominal
CHCOCNC1	Melanoma or other cancer	Nominal

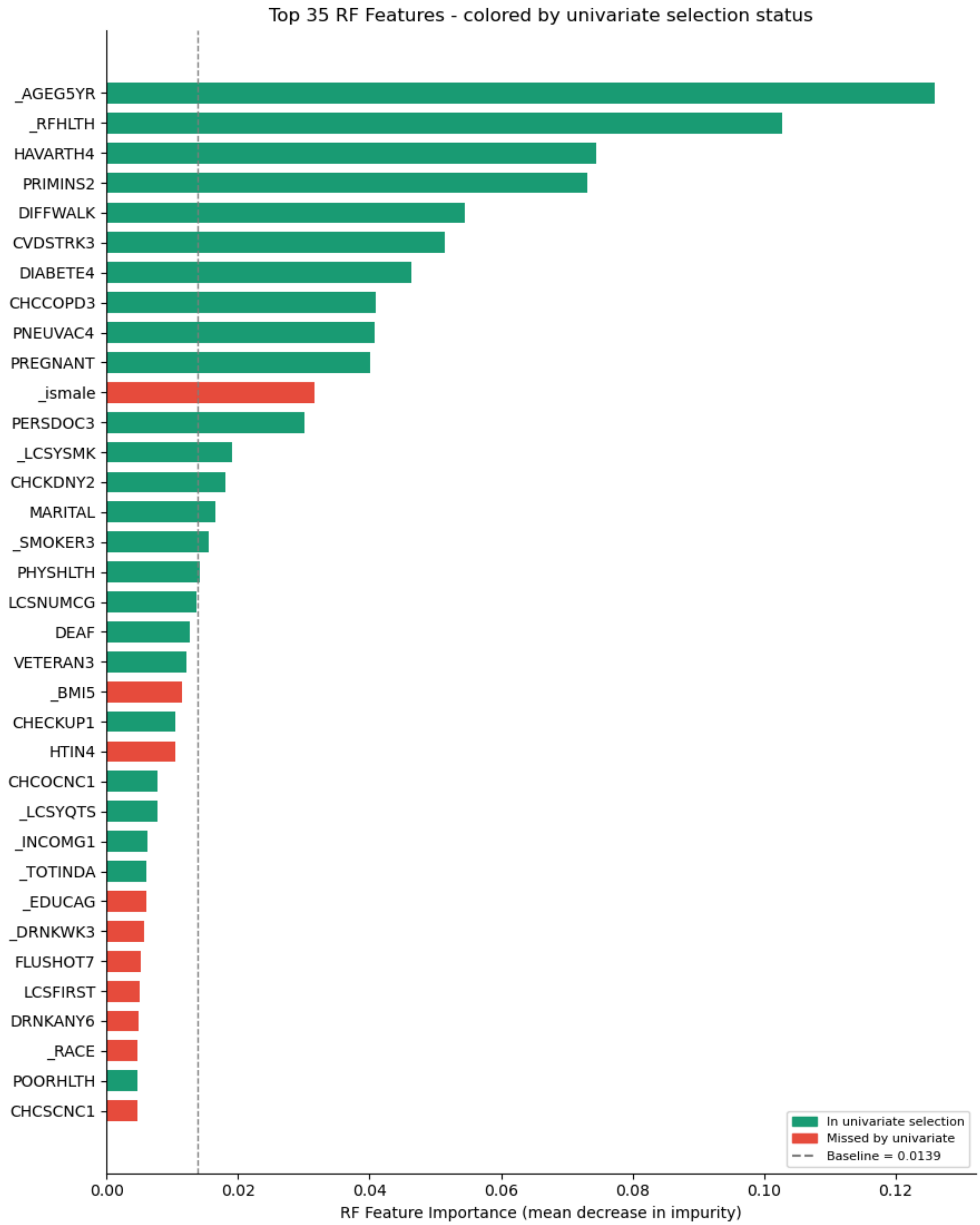
Feature	Description	Type
LSATISFY	Life satisfaction	Ordinal
SDLONELY	Loneliness frequency	Ordinal
DEAF	Serious hearing difficulty	Nominal
BLIND	Serious vision difficulty	Nominal
DECIDE	Difficulty concentrating	Nominal
DIFFWALK	Difficulty walking/climbing	Nominal
DIFFDRES	Difficulty dressing/bathing	Nominal
DIFFALON	Difficulty doing errands alone	Nominal
_HLTHPL2	Has health insurance	Nominal
PERSDOC3	Has personal doctor	Nominal
MEDCOST1	Couldn't afford doctor	Nominal
CHECKUP1	Time since last checkup	Ordinal
FLUSHOT7	Flu shot	Nominal
PNEUVAC4	Pneumonia vaccine	Nominal
SSBSUGR2	Daily sugary drink frequency	Continuous
ACEDEPRS	Lived with depressed person	Nominal
ACEDRINK	Lived with problem drinker	Nominal
ACEDRUGS	Lived with drug user	Nominal
ACEPRISN	Lived with incarcerated person	Nominal
ACEDIVRC	Parents divorced/separated	Nominal
ACEPUNCH	Witnessed domestic violence	Ordinal
ACEHURT1	Physically hurt by parent	Ordinal
ACEADSAF	Had protective adult	Ordinal

Feature	Description	Type
CIMEMLO1	Worsening memory/thinking	Nominal
CDSOCIA1	Cognitive difficulty affecting social life	Nominal
_BMI5	Continuous BMI	Continuous
HTIN4	Height in inches	Continuous
MSCODE	Metropolitan status	Nominal

Appendix B: Selected Features based on Univariate Analysis (27 Total)



Appendix C: Univariate vs. Random Forest Feature Selection



Appendix D: Multicollinearity Check (0 Features above 0.7)